



Qualcomm's Snapdragon 8 Gen2

newer device categories in the near future, like AR/VR headsets or smart wearables. In response, Patrick highlights an interesting perspective.

"In terms of the kind of innovation happening in the smartphone industry, we actually don't see it slowing down anytime soon. I think on the form factor side, many devices do look somewhat similar, especially the non-foldable devices. So yes, the range of different form factors is reducing – that's true. But we're very excited about the advent of foldable devices," explains Patrick, pointing out how both he and his wife use different foldable smartphones. They open a whole set of new experiences for users to dive into, and more OEMs will continue to experiment with foldable devices, unlocking new ways of doing things smarter and better than ever before, according to Patrick.

"When we look at camera solutions, we look at things like optical image stabilisation and full zoom lenses, so we are seeing a lot of different kinds of innovation in the handset business.

a strong presence in automotive for many years. We have wearable solutions, headsets, etc. In this broader ecosystem of smart wearable devices, we see the phone playing a very important role in enabling all these future experiences," says Patrick.

This is because for many people the phone is the one device that will always be with you – an absolutely indispensable device that's always gonna be with everyone, emphasises Patrick. "The smartphone will act as a natural hub to enable a personal cloud of devices, like the electronics headset, hopefully AR goggles one day, as well as a myriad of wearable devices on your person. A whole class of innovation and new user experiences will emerge with these different devices working together. And we're excited from the smartphone team to be a big part of that journey, no doubt about that," he says.

No doubt all the new innovative device form factors are great, but fundamentally all of them will still need to be powered by batteries. We asked Patrick why innovations in battery tech haven't kept pace with chip-level or form factor innovations; why battery efficiency continues to be a problem that needs to be solved indirectly – how everything that interfaces with the battery in a device becomes more efficient than the battery itself.

"It's a fair question and I probably don't have a good response honestly,"

understand your question and I share your frustration that we can't make it easier somehow by radically improving battery charge density in some way."

Patrick suggests there's certainly been some improvements in battery tech, "but I agree, there's been slow and linear improvements compared to the kind of exponential improvement we've had on the semiconductor side of the battery equation. From Qualcomm's perspective, we work very

hard on the technologies around the battery, on things like battery charging, operating at lower and lower voltages so we can use more of that battery range. But I agree, the industry at large hasn't made big strides on disrupting charge density within that battery."

With the new year 2023 just around the corner, what does

Qualcomm think about

increased competition from the likes of MediaTek and Intel going forward? Here's what Chris Patrick says about Qualcomm's drive for innovation.

"Competition's definitely not new to Qualcomm. We've always had strong competitors for 35+ years. I've been at Qualcomm for 26+ years, and I think competition gives us our edge, giving us the incentive to push, compete and come up with the very best products. When we think about MediaTek, Intel or anyone else in the industry, I think it's great to have them as part of the ecosystem, but we really focus on ourselves. At Qualcomm, we have this very deep technology pipeline, several sets of different topics that we've been researching. We're always very confident in the depth of the technology pipeline, with what we can bring to the industry. Despite the vigorous competition from MediaTek or others, we're never worried about our position in the industry. We're just always worried about being motivated to push ourselves and keep driving that pace of innovation forward," concludes Patrick. ■



31% Qualcomm held 33 per cent market share in the global smartphone chipset market in Q3 2022

On goggles, Qualcomm is working very hard with the Snapchat family of devices, different VR devices, AR devices. We're expanding into compute in a big way, we've always had

Patrick admits, while responding, "as a Qualcomm spokesperson, I can tell you we're working very hard to find ways to keep inching down a device's power consumption. I'm with you, I